The Romer Model

3.5 The Romer Model

Overview

- The model of **P. Romer (1990)** of **endogenous technological change**.
- In this model, R&D is under taken by **profit-maximizing economic factors**.
- R&D fuels growth
  - which in turn affects the incentives for devoting resources to R&D.
- Any model where the creation of knowledge is motivated by the returns
  - the knowledge commands in the market must involve **departures from perfect competition**

- If knowledge is sold at marginal cost
  - the creators of knowledge earn negative profits.
- Romer deals with this issue by assuming that
  - knowledge consists of distinct ideas and
  - Inputs into production that embody different ideas are imperfect substitutes.
  - He also assumes that the developer of an idea has monopoly rights to the use of the idea.

- These assumptions imply that
  - the developer can charge a price above marginal cost for the use of his or her idea.
  - The resulting profits provide the incentives for R&D.

- The reason for constructing the model this way is
  - it simplifies the analysis dramatically.
  - Models of this type exhibit **no transition dynamics**.
  - In response to a shock, the economy jumps immediately to its new BGP.

- Two types of simplifications of the model
- The **first** are assumptions about **functional forms** **and** **parameter values**
  - The model in P. Romer (1986) that launched new growth theory is closely related to our **learning-by-doing model with $\varphi = 1$ and $n = 0$**
    (analogous to the assumptions of $\theta = 1$ and $n = 0$ in R&D Model without capital**)**
- The other is the **elimination of all types of physical and human capital**

The Ethier Production Function and the Returns to Knowledge Creation

- The first step in presenting the model is to describe
  - how knowledge creators have market power.
- Thus for the moment
  - we take the **level of knowledge** as **given** and
  - describe how **inputs embodying different ideas** **combine to produce final output**.

- We assume that there is a **range of ideas** that are currently available that extends from **0 to A**, where A > 0
- We will use ***L(i)*** to denote both
  - the **quantity of labor** devoted to **producing input i** and
  - the **quantity of input i** that **goes into final-goods production**.

For ideas that have not yet been discovered (that is, for i >A), inputs embodying the ideas cannot be produced at any cost.

The specific assumption about how the inputs combine to produce final output uses the **production function proposed by Ethier (1982):**

$$Y = \left[\int_{i=0}^{A} L(i)^{\varphi} di\right]^{\frac{1}{\varphi}} \qquad 0 < \varphi < 1 \qquad (3.28)$$

Where, φ = substitutability among inputs

To see the implications of this function,

- let L_Y denote the **total number of workers producing inputs**, and
- suppose the number producing each available input is the same.
- Then $L(i) = \frac{L_Y}{A}$ for all i, and so

$$Y = \left[A\left(\frac{L_Y}{A}\right)^{\varphi}\right]^{\frac{1}{\varphi}}$$

$$= A^{\frac{1-\varphi}{\varphi}} L_Y \qquad (3.29)$$

Explanation:

$$Y = [\int_{i=0}^{A} L(i)^{\varphi} di]^{\frac{1}{\varphi}}$$

$$Y = [\int_{i=0}^{A} (\frac{L_Y}{A})^{\varphi} di]^{\frac{1}{\varphi}}$$

$$Y = [(\frac{L_Y}{A})^{\varphi} \int_{i=0}^{A} di]^{\frac{1}{\varphi}}$$

$$Y = [[(\frac{L_Y}{A})^{\varphi}. i]_0^A]^{\frac{1}{\varphi}}$$

$$\mathrm{Y} = [(\frac{L_Y}{A})^{\varphi}.\mathrm{A} - (\frac{L_Y}{A})^{\varphi}.0]^{\frac{1}{\varphi}}$$

$$\mathrm{Y} = [\mathrm{A}(\frac{L_Y}{A})^{\varphi}]^{\frac{1}{\varphi}}$$

$$\mathrm{Y} = [\mathrm{A}\frac{L_Y{}^{\varphi}}{A^{\varphi}}]^{\frac{1}{\varphi}}$$

$$\mathrm{Y} = [A^{(1-\varphi)} L_Y{}^{\varphi}]^{\frac{1}{\varphi}}$$

$$\mathrm{Y} = A^{\frac{1-\varphi}{\varphi}} L_Y{}^{\frac{\varphi}{\varphi}}$$

$$\mathrm{Y} = A^{\frac{1-\varphi}{\varphi}} L_Y$$

- This expression has **two critical implications**.
  - First, there are **constant returns to L_Y**
  - Second, output is **increasing in A**

- Consider **the cost-minimization problem** of a representative output producer.
- Let ***p(i)*** denote **the price charged**
  - by the holder of the patent on idea i
  - for each unit of the input embodying that idea.

The Lagrangian for the problem of producing one unit of output at minimum cost is

$$\mathcal{L} = \int_{i=0}^{A} p(i)L(i)di - \lambda\{[\int_{i=0}^{A} L(i)^{\varphi} di]^{\frac{1}{\varphi}} - 1\} \qquad (3.30)$$

Explanation:

Minimize: $\int_{i=0}^{A} p(i)L(i)di$

Subject to: $[\int_{i=0}^{A} L(i)^{\varphi}di]^{\frac{1}{\varphi}} = 1$

Therefore, the lagrangian is:

$\mathcal{L} = \int_{i=0}^{A} p(i)L(i)di + \lambda\,\{1 - \ [\int_{i=0}^{A} L(i)^{\varphi}di]^{\frac{1}{\varphi}}\}$

$\mathcal{L} = \int_{i=0}^{A} p(i)L(i)di - \ \lambda\,\{[\int_{i=0}^{A} L(i)^{\varphi}di]^{\frac{1}{\varphi}} - 1\}$

The FOC for an individual *L(i)* is

$$p(i) = \lambda L(i)^{\varphi-1} \tag{3.31}$$

Explanation:

According to FOC

$\frac{\delta\mathcal{L}}{\delta\, L(i)} = \frac{\delta}{\delta\, L(i)}\,(\int_{i=0}^{A} p(i)L(i)di - \ \lambda\,\{[\int_{i=0}^{A} L(i)^{\varphi}di]^{\frac{1}{\varphi}} - 1\}) = 0$

$\Rightarrow \frac{\delta}{\delta\, L(i)} \int_{i=0}^{A} p(i)L(i)di - \frac{\delta}{\delta\, L(i)}\ \lambda\,\{[\int_{i=0}^{A} L(i)^{\varphi}di]^{\frac{1}{\varphi}} - 1\} = 0$

$\Rightarrow\ p(i) - \lambda\frac{1}{\varphi}\,[\int_{i=0}^{A} L(i)^{\varphi}di]^{\frac{1}{\varphi}-1}\ \varphi L(i)^{\varphi-1} = 0$

$\Rightarrow\ p(i) - \lambda L(i)^{\varphi-1} = 0$ [since $[\int_{i=0}^{A} L(i)^{\varphi}di]^{\frac{1}{\varphi}} = 1$]

$\Rightarrow\ p(i)\ = \lambda L(i)^{\varphi-1}$

Equation (3.31) implies $L(i)^{\varphi-1} = \frac{p(i)}{\lambda}$ which in turn implies

$$L(i) = [\frac{p(i)}{\lambda}]^{\frac{1}{\varphi-1}}$$

$$= [\frac{\lambda}{p(i)}]^{\frac{1}{1-\varphi}} \tag{3.32}$$

- Equation (3.32) shows that
  - the holder of the patent on an idea faces a downward-sloping demand curve
  - ***L(i)*** is a smoothly **decreasing** function of ***p(i)***
  - When φ is **closer to 1**
    - the marginal product of an input declines more slowly as the quantity of the input rises.
    - As a result, the inputs are closer substitutes, and so
    - the **elasticity of demand for each input is greater**

- Because
  - firms **producing final output** face **constant costs** for **each input** and
  - **the production function** exhibits **constant returns**,
    - **marginal cost equals average cost**
  - As a result, **these firms earn zero profits**.

The Rest of the Model

The remainder of the model involves **four sets of assumptions**.

i) *The first set of assumption:*

The first set concern **economic aggregates**.

- **Population is fixed** and equal to $L > 0$
- We let $L_A\ (t)$ denote the number of workers **engaged in R&D** at time t
- And $L_Y\ (t) = \int_{i=0}^{A(t)} L(i)^{\varphi} di$ is the total number of workers **producing inputs** at time t
- then **equilibrium** in the labor market at *t* requires

$$L_A\ (t) + L_Y\ (t) = \bar{L} \tag{3.33}$$

- The **production function** for **new ideas** is
  - **linear in L_A** (the number of workers employed in R&D)
  - **proportional to A** (the existing stock of knowledge)

$$\dot{A}(t) = \mathrm{B}L_A\ (t)\ A(t) \qquad\qquad B > 0 \tag{3.34}$$

  - the initial level of A, A(0), is assumed to be strictly positive.

- These assumptions are chosen to give the model the **aggregate dynamics** of a **linear growth model**.

ii) *The second set of assumption:*

The second group of assumptions concern **the microeconomics of household behavior**.

- The behavior is **similar** to that of **RCK economy**
  - Individuals are infinitely lived
  - maximize a conventional utility function
  - Individuals' discount rate is ρ

- their instantaneous utility function is logarithmic.
- the representative individual's **lifetime utility** is

$$U = \int_{t=0}^{\infty} e^{-\rho t} lnC(t)\, dt \qquad \rho > 0 \tag{3.35}$$

- As in the RCK model, the individual's **budget constraint** is

$$\int_{t=0}^{\infty} e^{-r(t)} c(t)\, dt \leq X(0) + \int_{t=0}^{\infty} e^{-r(t)} w(t)\, dt \tag{3.36}$$

Where, *r* is the interest rate

X(0) is initial wealth per person, and

w(t) is the wage at t

iii) *The third set of assumption:*

The third set of assumptions concern **the microeconomics of R&D**.

- There is **free entry into idea creation**
  - anyone can hire $\frac{1}{BA(t)}$ units of labor at the prevailing wage *w(t)*
  - and produce a new idea (see [3.34])
  - the cost of creating that idea will be $\frac{w(t)}{BA(t)}$

- The creator of an idea is granted **permanent patent rights** to the use of the idea

- The free-entry condition in R&D requires that
  - the **present value of the profits** earned from selling the **input embodying an idea** equals the **cost of creating it**.
- If idea ***i*** is created at time ***t***
  - and let ***π(i,τ)*** denote the **profits** earned by the creator of the idea at time ***τ***
  - Then this condition is

$$\int_{\tau=t}^{\infty} e^{-r(\tau-t)}) \pi(i,\tau) d\tau = \frac{w(t)}{BA(t)} \tag{3.37}$$

iv) *The fourth set of assumption:*

The final assumptions of the model concern **general equilibrium**

- **First,** the assumption that the labor market is competitive
  - implies that the wage paid in R&D and the wages paid by all input producers are equal.
- **Second,** the only asset in the economy is the patents.
- **Finally,** the only use of the output good is for consumption
  - Equilibrium in the goods market at time t requires

$$C(t)\, \bar{\bar{L}} = Y(t) \tag{3.38}$$

Solving the Model

The problem of a patent holder

Price of input

- The first step in solving the model is to consider **the problem of a patent holder**
  - **choosing the price** to charge for his or her input at a point in time.
- A standard result from microeconomics is that
  - the **profit-maximizing price** of a monopolist **is** $\frac{\eta}{\eta-1}$ **times marginal cost**
  - where η is the **elasticity of demand**
- In our case, **the elasticity of demand** is constant and equal to $\frac{1}{1-\varphi}$
- And the **marginal cost** at time t is ***w(t)***
- Each monopolist therefore charges (price)

$$\frac{\frac{1}{1-\varphi}}{\frac{1}{1-\varphi}-1}w(t),\ or\ \frac{w(t)}{\varphi}$$

Explanation:

$$\frac{\frac{1}{1-\varphi}}{\frac{1}{1-\varphi}-1}w(t)=\frac{\frac{1}{1-\varphi}}{\frac{1-(1-\varphi)}{1-\varphi}}w(t)$$

$$=\frac{\frac{1}{1-\varphi}}{\frac{\varphi}{1-\varphi}}w(t)$$

$$=\frac{1}{1-\varphi}\cdot\frac{1-\varphi}{\varphi}\cdot w(t)$$

$$=\frac{w(t)}{\varphi}$$

Patent-holder's profits

Knowing the price each monopolist charges

- allows us to determine his or her **profits** at a point in time.

- Given our assumption that
  - L_A is constant and

- the requirement that $L_A(t) + L_Y(t) = \bar{L}$
  The quantity of each input used at time t is

$$\frac{\bar{L}-L_A}{A(t)}$$

- **Each patent-holder's profits** are thus

$$\pi(t) = \frac{\bar{L}-L_A}{A(t)}\left[\frac{w(t)}{\varphi} - w(t)\right]$$

$$= \frac{1-\varphi}{\varphi}\frac{\bar{L}-L_A}{A(t)}w(t) \qquad (3.39)$$

Explanation:

Patent-holder's profit = Patent-holder's revenue − Patent-holder's cost

$$\pi(t) = \frac{\bar{L}-L_A}{A(t)}\frac{w(t)}{\varphi} - \frac{\bar{L}-L_A}{A(t)}w(t)$$

$$= \frac{\bar{L}-L_A}{A(t)}\left[\frac{w(t)}{\varphi} - w(t)\right]$$

$$= \frac{\bar{L}-L_A}{A(t)}\left[\frac{1}{\varphi} - 1\right]w(t)$$

$$= \frac{1-\varphi}{\varphi}\frac{\bar{L}-L_A}{A(t)}w(t)$$

Present value of profits

- To determine the present value of profits from an invention, and hence the incentive to innovate,
- we need to determine
  - **the economy's growth rate** and
  - **the interest rate**

Economy's growth rate

According to Equation (3.34) $\dot{A}(t) = BL_A(t)A(t)$

- This implies that
- if L_A is constant,
  - Growth rate of A is, $\frac{\dot{A}(t)}{A(t)} = BL_A$

According to Equation (3.29) $Y(t) = A(t)^{\frac{1-\varphi}{\varphi}} L_Y(t)$

- Since $L_Y(t)$ is constant
  - the growth rate of Y is $\frac{1-\varphi}{\varphi}$ times the growth rate of A (which is, BL_A)
    - Growth rate of Y is, $\frac{\dot{Y}(t)}{Y(t)} = \frac{1-\varphi}{\varphi} BL_A$

The interest rate

- Both **consumption** and the **wage grow at the same rate** as **output**.
- And the **growth rate of the wage equals** the **growth rate of output**
- Thus, **Growth rate of output** $=$ **Growth rate of consumption**

  - Once we know the growth rate of consumption, finding the **real interest rate** is straightforward.
  - Euler Equation of the RCK model can be written as:

$$\frac{\dot{C}(t)}{C(t)} = \frac{r(t)-\rho}{\theta}$$

    - With logarithmic utility, θ is 1

  - Thus equilibrium requires

$$r(t) = \rho + \frac{\dot{C}(t)}{C(t)}$$

$$= \rho + \frac{1-\varphi}{\varphi} BL_A \qquad (3.40)$$

  - **Thus if L_A is constant, the real interest rate is constant**

Explanation:

$$\frac{\dot{C}(t)}{C(t)} = \frac{r(t)-\rho}{\theta}$$

$$r(t) - \rho = \frac{\dot{C}(t)}{C(t)} \qquad \text{(when } \theta = 1 \text{)}$$

$$r(t) = \rho + \frac{\dot{C}(t)}{C(t)}$$

$$r(t) = \rho + \frac{1-\varphi}{\varphi} BL_A$$ (*Since,* Growth rate of output = Growth rate of consumption)

Present value of profits

- According to Equation (3.39) profits at *t* are $\frac{1-\varphi}{\varphi} \frac{\bar{L}-L_A}{A(t)}$ w(t)
- Growth rate of profit = Growth rate of w – Growth rate of A

$$= \frac{1-\varphi}{\varphi} BL_A - BL_A$$
$$= [\frac{1-\varphi}{\varphi} - 1] BL_A$$
$$= [\frac{1-\varphi-\varphi}{\varphi}] BL_A$$
$$= \frac{1-2\varphi}{\varphi} BL_A$$

- Profits are discounted at the interest rate r. (i.e., at rate $\rho + \frac{1-\varphi}{\varphi} BL_A$)

- The **present value of the profits** earned from the discovery of a new idea at time t is therefore

$$\text{PV of } \pi(t) = \frac{\frac{1-\varphi}{\varphi} \frac{\bar{L}-L_A}{A(t)} \mathrm{w(t)}}{\rho + \frac{1-\varphi}{\varphi} BL_A - \frac{1-2\varphi}{\varphi} BL_A}$$

$$= \frac{1-\varphi}{\varphi} \frac{\bar{L}-L_A}{\rho + BL_A} \frac{w(t)}{A(t)} \qquad (3.41)$$

Explanation:

Present value of profit at time t $= \frac{\text{profit at time t}}{\text{discount rate} - \text{growth rate of profit}}$

$$\text{PV of } \pi(t) = \frac{\frac{1-\varphi}{\varphi}\,\frac{\bar{L}-L_A}{A(t)}\,\mathrm{w(t)}}{\rho + \frac{1-\varphi}{\varphi}BL_A - \frac{1-2\varphi}{\varphi}BL_A}$$

$$= \frac{\frac{1-\varphi}{\varphi}\,\bar{L}-L_A\,\frac{\mathrm{w(t)}}{A(t)}}{\rho + BL_A\,[\frac{1-\varphi}{\varphi} - \frac{1-2\varphi}{\varphi}]}$$

$$= \frac{\frac{1-\varphi}{\varphi}\,\bar{L}-L_A\,\frac{\mathrm{w(t)}}{A(t)}}{\rho + BL_A\,[\frac{1-\varphi-1+2\varphi}{\varphi}]}$$

$$= \frac{\frac{1-\varphi}{\varphi}\,\bar{L}-L_A\,\frac{\mathrm{w(t)}}{A(t)}}{\rho + BL_A[\frac{\varphi}{\varphi}]}$$

$$= \frac{1-\varphi}{\varphi}\,\frac{\bar{L}-L_A}{\rho+BL_A}\,\frac{w(t)}{A(t)}$$

Equilibrium value of L_A

- If the amount of **R &D is strictly positive**
  - **the present value of profits** from an invention **must equal the costs of the invention**
- Since one worker can produce BA(t) ideas per unit time
  - the cost of an invention is $\frac{w(t)}{BA(t)}$
- The equilibrium condition is therefore

$$\frac{1-\varphi}{\varphi}\,\frac{\bar{L}-L_A}{\rho+BL_A}\,\frac{w(t)}{A(t)} = \frac{w(t)}{BA(t)} \qquad (3.42)$$

Solving this equation for L_A yields

$$L_A = (1-\varphi)\ \bar{L} - \frac{\varphi\rho}{B} \qquad (3.43)$$

Explanation:

$$\frac{1-\varphi}{\varphi}\frac{\bar{L}-L_A}{\rho+BL_A}\frac{w(t)}{A(t)} = \frac{w(t)}{BA(t)}$$

$$\Rightarrow \frac{1-\varphi}{\varphi}\frac{\bar{L}-L_A}{\rho+BL_A} = \frac{w(t)}{BA(t)} \cdot \frac{A(t)}{w(t)}$$

$$\Rightarrow \frac{1-\varphi}{\varphi}\frac{\bar{L}-L_A}{B\,(\frac{\rho}{B}+L_A)} = \frac{1}{B}$$

$$\Rightarrow \frac{1}{B}\frac{1-\varphi}{\varphi}\frac{\bar{L}-L_A}{(\frac{\rho}{B}+L_A)} = \frac{1}{B}$$

$$\Rightarrow \frac{1-\varphi}{\varphi}\frac{\bar{L}-L_A}{(\frac{\rho}{B}+L_A)} = 1$$

$$\Rightarrow (1-\varphi)\,(\bar{L}-L_A) = \varphi\,(\frac{\rho}{B}+L_A)$$

$$\Rightarrow (1-\varphi)\,\bar{L} - (1-\varphi)\,L_A = \frac{\varphi\rho}{B} + \varphi L_A$$

$$\Rightarrow (1-\varphi)\,\bar{L} - \frac{\varphi\rho}{B} = \varphi L_A + (1-\varphi)\,L_A$$

$$\Rightarrow (1-\varphi)\,\bar{L} - \frac{\varphi\rho}{B} = L_A\,(\varphi + 1 - \varphi)$$

$$\Rightarrow L_A = (1-\varphi)\,\bar{L} - \frac{\varphi\rho}{B}$$

- However, the amount of R&D need not be strictly positive.
- In particular, when (3.43) implies $L_A < 0$
  - R & D will be 0
- Thus we can modidy (3.43) to

$$L_A = max\,\{(1-\varphi)\,\bar{L} - \frac{\varphi\rho}{B},\, 0\} \tag{3.44}$$

Finally, since the growth rate of output is $\frac{1-\varphi}{\varphi} BL_A$, we have

$$\frac{\dot{Y}(t)}{Y(t)} = \max\,\{\frac{1-\varphi)^2}{\varphi} B\bar{L} - (1-\varphi)\,\rho,\, 0\} \tag{3.45}$$

- Thus we have succeeded in describing **how long-run grow this determined**
- Since none of the terms on the right-hand side of (3.43) are time-varying
  - **the equilibrium value of** L_A is **constant**

Explanation:

$$\frac{\dot{Y}(t)}{Y(t)} = \frac{1-\varphi}{\varphi}\ B L_A$$

$$= \frac{1-\varphi}{\varphi}\ B.\max\{(1-\varphi)\,\bar{L} - \frac{\varphi\rho}{B}, 0\}$$

$$= \max\{\frac{1-\varphi)^2}{\varphi}\mathrm{B}\,\bar{L} - \frac{1-\varphi}{\varphi}\,B\,\frac{\varphi\rho}{B}, 0\}$$

$$= \max\{\frac{1-\varphi)^2}{\varphi}\mathrm{B}\,\bar{L} - (1-\varphi)\,\rho, 0\}$$

Implications

The model has **two major** sets of implications

- ***The first set of implication:***

The first concern the **determinants of long-run growth**.

- **Four parameters** affect the **economy's growth rate**

First, when individuals are less patient

- that is, **when *ρ* is higher**
  - fewer workers engage in R&D (equation [3.44])
  - **growth is lower** (equation [3.45])

Second, an **increase** in substitutability among inputs ***(φ)*** also **reduces growth**

There are **two reasons**

i) **fewer** workers engage in R&D (equation [3.44])
ii) although a given amount of R&D translates into the same growth rate of A (equation [3.34]), **a given growth rate of A** translates into **slower output growth** (equation [3.29])

Third, an **increase in productivity** in the R&D sector ***(B)*** **increases** growth.

There are again **two effects** at work.

i) a rise in B raises growth for a given number of workers engaged in R&D.

ii) increased productivity in R&D draws more workers into that sector.

Finally, an **increase** in the size of the population ***(L)*** **raises** long-run growth.

There are **two effects**

i) growth increases for a given fraction of workers engaged in R&D, and the fraction of workers engaged in R&D

ii) an increase in the size of the economy expands the market an inventor can reach, and so increases the returns to R&D

❖ ***The second set of implication:***

- The model's second major set of implications concern
  - **the gap** between **equilibrium** and **optimal growth**
- Since the economy is not perfectly competitive, **there is no reason to expect the decentralized equilibrium to be socially optimal**.
- let us look for the constant level of L_A that yields the **highest level of lifetime utility** for the representative individual.
- Because all output is consumed
  - the representative individual's **consumption** is $\frac{1}{\bar{L}}$ **times output**
  - Equation (3.29) for output therefore implies that the representative individual's **consumption at time 0** is

$$C(0) = \frac{(\bar{L} - L_A) A(0)^{\frac{1-\varphi}{\varphi}}}{\bar{L}} \tag{3.46}$$

Explanation:

The representative individual's consumption $= \frac{1}{\bar{L}}$ times output

Therefore, $C(0) = \frac{1}{\bar{L}} \cdot Y(0)$

$$= \frac{A(0)^{\frac{1-\varphi}{\varphi}} L_Y}{\bar{L}} \qquad \text{[from (3.29)]}$$

$$= \frac{A(0)^{\frac{1-\varphi}{\varphi}} (\bar{L} - L_A)}{\bar{L}} \qquad \text{[from (3.33)]}$$

$$= \frac{(\bar{L} - L_A) A(0)^{\frac{1-\varphi}{\varphi}}}{\bar{L}}$$

- The representative individual's lifetime utility is therefore

$$U = \int_0^{\infty} e^{-\rho t} ln\, [\frac{(\bar{L} - L_A)}{\bar{L}} A(0)^{\frac{1-\varphi}{\varphi}}\, e^{[\frac{1-\varphi}{\varphi}]BL_A t}]dt \quad (3.47)$$

- One can show that the solution to this integral is

$$U = \frac{1}{\rho} \left(ln \frac{\bar{L} - L_A}{\bar{L}} + \frac{1-\varphi}{\varphi} ln\, A\,(0) + \frac{1-\varphi}{\varphi} \frac{BL_A}{\rho} \right) \quad (3.48)$$

- Maximizing this expression with respect to L_A shows that the socially optimal level of L_A is given by

$$L_A^{OPT} = \max\, \{\ \bar{L} - \frac{\varphi}{1-\varphi} \frac{\rho}{B}, 0\} \quad (3.49)$$

- Comparing this expression with equation(3.44)

$$L_A^{EQ} = (1-\varphi) L_A^{OPT} \quad (3.50)$$

- **If $\boldsymbol{\varphi > \frac{1}{2}}$**
  - **the profits** of existing patent-holders are **reduced** by an **increase in A**
- If $\varphi < \frac{1}{2}$
  - **the profits** of existing patent-holders are **raised** by an **increase in A**

- The **equilibrium** number of **workers** engaged in R&D is
  - always **less than** the **optimal number**
- Thus **growth** is always in efficiently **low**

- Moreover, the proportional **gap** between the equilibrium and optimal numbers (and hence between equilibrium and optimal growth)
  - depends only on a single parameter, φ
  - The smaller the degree of differentiation among inputs embodying different ideas (that is, the **greater is** φ), the **greater the gap**

Extensions

First, Introducing **physical capital** **does not change** the model's **central messages**.

- In Romer's model, where physical capital is not an input into R&D,
  - **policies that increase physical-capital investment** have only **level effects**, **not growth effects**.
- In variants **where capital enters the production function for ideas**, such policies generally have **growth effects**

Second,

Extending the model into a nonlinear version changes its implication.

- We have been considering correspond to a **linear growth model**.
- Jones (1995a) extends the Romer model to the case
  - where the **exponent on A** in the production function for ideas is **less than 1.**
  - This **creates transition dynamics**
  - More importantly, it **changes the model's messages** concerning t**he determinants of long-run growth**.
  - The macroeconomics of Jones's model correspond to those of a **semi-endogenous growth model**.
  - As a result, **long-run growth** depends only **on the rate of population growth**.

Third,

If technology improves existing inputs, the model **may identify other factors** that affect long-run economic performance.

- In Romer's model, technological progress takes the form of expansion of the number of inputs into production.
- An alternative is that it takes the form of improvements in existing inputs.
- This leads to the "**quality ladder**" models of Grossman and Helpman (1991a) and Aghion and Howitt (1992).
- Quality-ladder models do not produce sharply different answers than expanding-variety models concerning the long-run growth and level of income.
  - But they identify additional microeconomic determinants of incentives for innovation, and so
  - show **other factors that affect long-run economic performance**.

3.6 Empirical Application: Time-Series Tests of Endogenous Growth Models

Jones (1995b) raises a critical issue about these models: Does growth in fact vary with the factors identified by the models in the way the models predict?

Are growth rates stationary?

Jones examines two methods for testing the predictions of fully endogenous growth models regarding growth rate changes. He notes that these models suggest that changes in parameters can permanently alter growth rates, leading him to investigate whether the growth rate of income per person is stationary or nonstationary. Stationarity implies a constant distribution over time, while nonstationarity indicates that shocks to growth have lasting effects.

Jones conducts several statistical tests, including regression analyses and the augmented Dickey-Fuller test, to explore the stationarity of U.S. income growth data from 1880 to 1987. While his results indicate strong evidence for stationarity, he argues that **these findings do not provide meaningful insights into significant economic changes in growth**. The tests suggest that while growth appears stable in the short term, they do not address whether there have been significant long-term shifts. Ultimately, Jones concludes that **the tests of stationarity may not be relevant for understanding substantive economic questions about growth trends**.

The Magnitudes and Correlates of Changes in Long-Run Growth

Jones's second approach analyzes the relationships between growth determinants from endogenous growth models and actual growth rates. He first examines learning-by-doing models, noting that while variables like the output-capital ratio and depreciation are stable, investment rates are increasing. This leads the model to predict a growth increase of about 0.4 percentage points per decade, which contrasts sharply with observed trends in the U.S.

Next, he explores R&D-based endogenous growth models, which suggest that the significant increase in R&D investment and personnel should correlate with a substantial rise in growth rates. However, this predicted growth surge is not supported by the data.

Overall, Jones concludes that **despite models predicting rapid growth due to rising determinants, actual growth has remained flat over the past century, highlighting a significant disconnect between theory and empirical evidence.**

Discussion

Jones interprets his findings as evidence supporting semi-endogenous growth models, which suggest decreasing returns to produced factors, rather than fully endogenous growth models. While some subsequent research maintains constant or increasing returns and posits that R&D activity per sector drives growth—despite a growing number of sectors—Jones points out two main issues with this view.

First, alongside population growth, investment shares in R&D and physical/human capital have also been rising, making stagnant growth puzzling for these models. Second, the parameter restrictions needed to eliminate scale effects in these models seem arbitrary and overly stringent. With decreasing returns, increases in saving or R&D investment may lead to temporary growth boosts rather than sustained increases, indicating that major economies are not on a conventional balanced growth path.

Ultimately, Jones suggests that **while R&D can continue rising for a time, growth will likely slow down significantly in the absence of countervailing forces, as these upward trends cannot persist indefinitely.**

3.7 Empirical Application: Population Growth and Technological Change since 1 Million B.C.

Kremer (1993) applies endogenous knowledge accumulation models to analyze historical dynamics of population, technology, and income. He argues that these models predict technological progress increases with population size, as a larger population leads to more discoveries and faster knowledge accumulation. Historically, this technological progress primarily resulted in population growth rather than significant increases in output per person. Before the Industrial Revolution, population expanded significantly while incomes remained near subsistence levels, limiting per capita output growth. Only in recent centuries has technological advancement started to substantially impact income levels. Thus, Kremer concludes that these models suggest a rising rate of population growth throughout most of human history.

A simple model

Kremer's model is a variation of existing models of endogenous growth, structured around three key equations.

First, it describes output (Y) as a function of technology (A), labor (L), and land (T):

Second, knowledge accumulation is modeled as being proportional to population and the stock of knowledge:

Third, population adjusts to maintain output per person at a subsistence level (y):

The model integrates a Malthusian assumption for population determination. To analyze it, Kremer first derives the population size that can be supported by a fixed land stock, showing that population is inversely related to the subsistence output level and directly related to technology and land.

Next, he examines the dynamics of technology and population growth. The model implies that population growth is tied to the growth rate of knowledge, leading to increasing population growth rates as long as certain conditions on parameters (α and θ) hold. Even with diminishing returns to knowledge production, the interplay between rising technology and population fosters ongoing growth, indicating a significant potential for expansion in population-driven technological advancement.

Results

Kremer tests his model's predictions about population growth using extensive population estimates dating back to 1 million B.C., finding a strong, linear relationship between initial population levels and growth rates. His regression analysis reveals a significant association, with higher populations correlating with faster growth.

He further examines the implications of isolation among regions over time, arguing that technological progress would be faster in areas with larger populations, given that these regions started with similar technology levels. Following the reestablishment of contact around 1500, he predicts that regions with larger populations would have higher population densities.

The data supports this, showing that in 1500, Eurasia-Africa had a population density of about 4.9 people per square kilometer, while the Americas, Australia, and Tasmania had significantly lower densities (0.4, 0.03, and 0.03, respectively), aligning with Kremer's model that links population size with technological advancement.

Discussion

Kremer's model, which posits that the rate of knowledge accumulation increases with population size, is confirmed by both time-series and cross-sectional data. This idea contrasts with traditional models like Solow, Ramsey, and Diamond, which assume exogenous innovation. While Kremer's findings, such as the proportionality of population growth to population levels, may seem intuitive, they underscore the complexity of accurately modeling technological progress over millennia.

The significance of Kremer's evidence lies not in pinpointing exact parameters like θ, but in utilizing growth theory to gain insights into key historical dynamics, particularly regarding long-term population trends and technological disparities between regions before 1500. His work demonstrates that new growth theory provides valuable understanding of these historical phenomena.

Population Growth versus Growth in Income per Person over the Very Long Run

Kremer highlights that throughout most of history, technological progress primarily resulted in population growth rather than increases in average income. However, in the modern era, technological advancements have led to both significant population growth and substantial increases in average income. This shift doesn't necessitate complex demographic changes, such as the adoption of contraception; instead, Kremer argues that Malthusian dynamics operate over time.

In his model, population growth is an increasing function of income at lower income levels, which means that as income rises, population growth also rises, tending to push income back down. When technological progress is slow, this adjustment keeps income per person close to subsistence levels. However, with rapid population growth, the impact of technological progress begins to enhance average income rather than just population size.

Kremer extends this idea by suggesting that once average income exceeds a certain threshold, population growth may decrease as income increases. This modification implies that population growth can peak and eventually decline, reinforcing the trend where technological progress increasingly benefits average income. If population growth becomes negative at high income levels, the model predicts that both population and technological progress may ultimately converge to zero.

3.8 Models of Knowledge Accumulation and the Central Questions of Growth Theory

The analysis of economic growth focuses on two key issues: the evolution of living standards over time and the disparities in income across different regions. Researchers initially hoped that models of R&D and knowledge accumulation would provide a unified explanation for global growth and income differences. However, the nonrivalry of knowledge presents a fundamental challenge, as knowledge used by one producer can also be utilized by others, suggesting that poor countries could adopt the same knowledge as rich countries.

While access to knowledge and technology can facilitate growth in poorer nations, issues like property rights and the ability to effectively use technology are significant barriers. Even in stable political environments, income levels do not automatically rise to match those of industrialized nations, indicating that other factors play a crucial role in economic disparities.

Moreover, the slow diffusion of technology and the complexities surrounding its utilization suggest that income differences stem from underlying factors beyond mere access to knowledge. Future research should focus on understanding these factors to better explain income disparities.

In contrast, the relevance of knowledge accumulation models is stronger regarding global growth, as the increase in knowledge is a key driver of higher output and living standards today compared to the past. While current models provide insights into the determinants and impacts of knowledge accumulation, they still require refinement for precise quantitative predictions related to policy interventions, like increasing funding for scientific research. Overall, the analysis indicates promising directions for further research into the relationship between knowledge and economic growth.